

Daniel Gross

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EDUCATION

University of Massachusetts Boston

Sep. 2023 - May. 2027

Bachelor of Arts in Computer Science

IDS Student Research Fellow @ MPSYCH Lab

September 2024 – May 2025

RESEARCH

SuperDeepfakeDetector.com | *JavaScript, TensorFlow.js, HTML/CSS*

February 2025 – May 2025

- Built a website which analyzes if a video is a Deepfake or not.
- Uses TensorFlow.js to run Deepfake Detection Models in the browser.
- Trained Convolutional Neural Networks (CNNs) for Deepfake Detection using TensorFlow with Python.
- Provides a quick way for anyone to determine the legitimacy of a video without any paywall.

SuperDeepfakeDetector V1 | *Python, PyTorch, Flask, HTML/CSS*

November 2024 – December 2024

- Implemented a website which determines whether or not a video is a Deepfake or not.
- Uses DeepfakeBench on the backend, computing the average accuracy of the detectors given a video file.
- Provides a quick way for anyone to determine the legitimacy of a video without any commitment.
- Aims to be a platform for researchers to test deepfake detector models on the client-side.

PROJECTS

AutoReducer.com | *JavaScript, HTML/CSS*

September 2025 – October 2025

- Developed a web application to automatically prepare pdf documents for 3-Hole printing.
- Implemented a completely clientside solution, to ensure 100% privacy of data.
- Resizes several document sizes to US Letter and prepares them for printing.
- Speeds up the workflow by automating the manual check of each document.

Fitness Web App | *Python, Flask, SQLite, HTML/CSS*

March 2024 – May 2024

- Developed a full-stack web application using Flask, with simple HTML/CSS as the frontend.
- Implemented a simple user login/signup feature using POST and GET requests to send data to the backend.
- Designed a simple food nutrition search, using a third party API to retrieve information about the queried item.
- Collaborated with other group members using AGILE methodologies and S.M.A.R.T. goals to speed development.

RELEVANT COURSEWORK

Implementing and Securing Large Language Models | *CS478*

Fall 2024

- Learned about the architecture of LLMs, tokenizers, transformers, and models such as GPT and BERT.
- Implemented a GPT model with PyTorch, utilizing a BPE tokenizer and AdamW optimizer.
- Conducted training on Nvidia A100 GPUs, tweaking and optimizing the model.
- Researched common adversarial attacks on LLMs, and the strategies employed to mitigate them.

Intermediate Computing with Data Structures | *CS210*

Spring 2024

- Studied the design and implementations of data structures and algorithms in Java.
- Designed various data structures, including variations of Heaps, Binary Trees, Hashmaps, and Graphs.
- Implemented several algorithms, such as Binary Search, Quick Sort, Dijkstra's Shortest Path, A* Search.
- Familiarized with the concept of Object-Oriented Programming with Java.

TECHNICAL SKILLS

Languages: JavaScript, Python, Java, HTML/CSS, Bash

Libraries: TensorFlow.js, PyTorch, Flask, Matplotlib, Numpy, Pandas

Developer Tools: Linux, Vim, Git